

Question 1: todo

Question 2: For the following proofs, let R be a commutative ring and $x, y, z \in R$.

- (a) *Proof.* We show that $(-x)(-y) + (-xy) = 0$, and thus $(-x)(-y) = xy$. As was shown in Proposition 5.7 in the notes, $-(xy) = (-x)y$. So, we compute that

$$(-x)(-y) + (-xy) = (-x)(-y) + (-x)y = (-x)(-y + y) = (-x) \cdot 0 = 0.$$

Thus, we are done. $\square$

- (b) *Proof.* Considering $x(y - z) = x(y + (-z))$, we can appeal to the distributive property of commutative rings. So, $x(y + (-z)) = xy + x(-z)$. Appealing to Proposition 5.7 in the notes, we have that $x(-z) = -(xz)$, we get that $xy + x(-z) = xy - xz$. $\square$

- (c) *Proof.* Considering $-1(x)$, we can appeal to Proposition 5.7 in the notes, and rewrite the expression as $-(1x)$. Appealing to R 's multiplicative identity, which is 1, we know $1x = x, \forall x \in R$. Thus, $-(1x) = -x$. $\square$

Question 3: *Proof.* Suppose R is an ordered ring, $x, y, z \in R, x < y, 0 < z$. It follows that $y - x \in R^+$ and $z \in R^+$. Appealing to the closure of positives, $(y - x)(z) \in R^+$. Distributing, we get that $yz - xz \in R^+$. Thus, $xz < yz$. $\square$

Question 4: q4

Question 5: *Proof.* We proceed by induction.

BC When $n = 0$, we have that $\sum_{k=0}^0 k^2 = 0 = \frac{n(n+1)(2n+1)}{6} \Big|_{n=0}$. So, the base case is true.

IS Assume $\sum_{k=0}^n k^2 = \frac{n(n+1)(2n+1)}{6}$. Consider $\sum_{k=0}^{n+1} k^2$. We can rewrite this as

$$\sum_{k=0}^{n+1} k^2 = \left(\sum_{k=0}^n k^2 \right) + (n+1)^2 = \frac{n(n+1)(2n+1)}{6} + (n+1)^2.$$

Rearranging, we find that this is equivalent to $\frac{2n^3 + 9n^2 + 13n + 6}{6}$. Thus,

$$\left(\sum_{k=0}^n k^2 \right) + (n+1)^2 = \frac{2n^3 + 9n^2 + 13n + 6}{6}.$$

We find that this is further equivalent to $\frac{(n+1)(n+2)(2n+1)}{6}$. Thus,

Thus, $\sum_{k=0}^0 k^2 = \frac{n(n+1)(2n+1)}{6}$ for all $n \in \mathbb{N}$. $\square$

Question 6: *Proof.* We proceed by induction.

BC When $n = 1$, we have that $\sum_{k=0}^1 = r + 1 = \frac{r^{(1+1)} - 1}{r - 1}$.

IS

□

Question 7: We claim that $\sum_{k=1}^n (2k - 1) = n^2, \forall n \in \mathbb{Z}$.

Proof. Summation is a linear operator, so $\sum_{k=1}^n (2k - 1) = 2 \sum_{k=1}^n k - \sum_{k=1}^n 1$. The first term is a known result from class: $\sum_{k=1}^n k = \frac{n(n+1)}{2}$. The second term is simply n . So, we get that $\sum_{k=1}^n (2k - 1) = n(n+1) - n = n^2$. Ask about in office hours if this is sufficient rigor! □

Question 8: We claim that $\sum_{k=1}^n \frac{1}{k(k+1)} = \frac{n}{n+1}, \forall n \in \mathbb{Z}$. We provide two different proofs.

Proof. First, we can proceed via algebraic manipulation. We find that $\frac{1}{k(k+1)} = \frac{1}{k} - \frac{1}{k+1}$. Rewriting, we find that the sum of these terms is a basic telescopic series:

$$\sum_{k=1}^n \left(\frac{1}{k} - \frac{1}{k+1} \right) = \left(\frac{1}{1} - \frac{1}{2} \right) + \left(\frac{1}{2} - \frac{1}{3} \right) + \left(\frac{1}{3} - \frac{1}{4} \right) \cdots \left(\frac{1}{n} - \frac{1}{n+1} \right).$$

We see that every term cancels with its opposite, other than 1 and $\frac{1}{n+1}$. Thus, $\sum_{k=1}^n \left(\frac{1}{k} - \frac{1}{k+1} \right) = 1 - \frac{1}{n+1} = \frac{n}{n+1}$. □

Proof. Another method of proof is by induction.

BC When $n = 1$, we have that $\sum_{k=1}^1 = \frac{1}{2} = \frac{n}{n+1} \Big|_{n=1}$.

IS Assume that $\sum_{k=1}^n \frac{1}{k(k+1)} = \frac{n}{n+1}$. Consider $\sum_{k=1}^{n+1} \frac{1}{k(k+1)}$. This can be rewritten:

$$\sum_{k=1}^{n+1} \frac{1}{k(k+1)} = \sum_{k=1}^n \frac{1}{k(k+1)} + \frac{1}{(n+1)(n+2)} = \frac{(n+1)^2}{(n+1)(n+2)}.$$

Simplifying, we see that $\frac{(n+1)^2}{(n+1)(n+2)} = \frac{n+1}{n+2}$. So, $\sum_{k=1}^{n+1} \frac{1}{k(k+1)} = \frac{n+1}{n+2}$. Thus, we are done. □